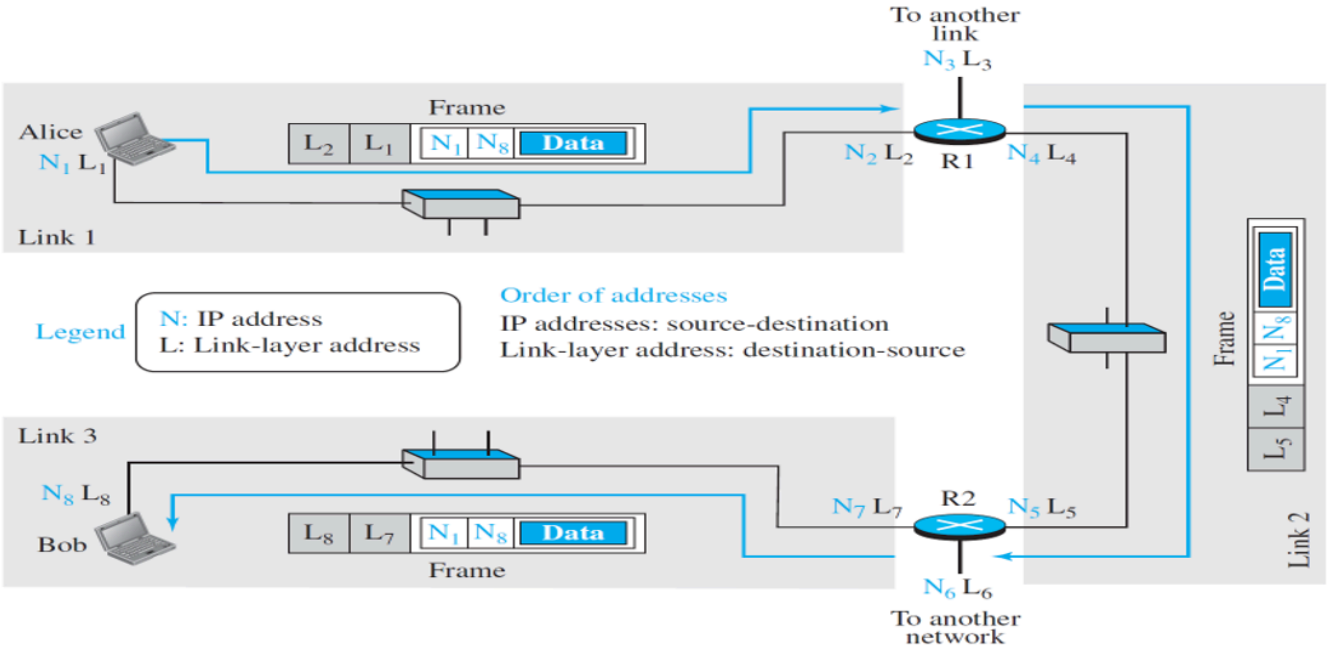
Status	Finished
Started	Thursday, 15 October 2026, 3:31 PM
Completed	Thursday, 15 October 2026, 3:40 PM
Duration	9 mins 9 secs

Question 1

Complete



Which statement about the addresses in the figure above is **NOT correct**?

- a. The source and destination IP addresses remain N_1 and N_8 as the packet travels from Alice to Bob.
- b. The destination link-layer address remains L_8 from Alice to Bob because the link-layer destination should always identify the final host.
- c. On Link 2, router R1 creates a new frame using L_4 as the source link-layer address and L_5 as the destination link-layer address.
- d. On Link 1, the frame uses Alice's link-layer address L_1 as the source and router R1's link-layer address L_2 as the destination.
- e. On Link 3, router R2 creates another new frame using L_7 as the source and Bob's link-layer address L_8 as the destination.

Question 2

Complete

Which statement about unicast, multicast, and broadcast addressing is **NOT correct**?

- a. A **multicast address** identifies a group of receivers that are members of the same multicast group.
- b. The Ethernet broadcast address is **FF:FF:FF:FF:FF:FF** and is intended for all devices in the local broadcast domain.
- c. An **ARP Request** is commonly sent using broadcast, while an **ARP Reply** is typically sent as unicast.
- d. A **unicast address** identifies one specific node or destination.
- e. A **multicast address** is used when a sender wants every device on the local network to receive the frame.

Question 3

Complete

Which statement about checking MAC and IP addresses is **NOT correct**?

- a. On Windows, `ipconfig /all` can be used to display both the IPv4 address and the Physical Address of a network adapter.
- b. On Linux, commands such as `ip a` or `ifconfig` can be used to inspect the IP address assigned to a network interface.
- c. On Linux, commands such as `ip link show` can be used to inspect the MAC address of a network interface.
- d. On mobile devices, the MAC address and IP address are usually found in network or device information settings.
- e. A MAC address and an IP address are the same identifier, so checking either one gives the same address value for the device.

Question 4

Complete

Which sequence correctly represents the basic **DHCP DORA** process?

Answer: Discover → Offer → Request → ACK

Question 5

Complete

Which statement about DHCP is **NOT correct**?

- a. DHCP can resolve a known IP address into the corresponding MAC address.
- b. DHCP can provide network configuration information to a host automatically.
- c. DHCP can provide a subnet mask and default gateway to a host.
- d. DHCP was developed from BOOTP to support automatic host configuration.
- e. DHCP can assign an IP address to a newly connected host automatically.

Question 6

Complete

Match each DHCP message with the correct description.

- DHCP Discover — The client requests the offered IP address from the selected DHCP server.
- DHCP Offer — A DHCP server offers an IP address and network configuration to the client.
- DHCP Request — The client asks for the MAC address of the default gateway.
- DHCP ACK — The DHCP server confirms the lease and configuration assigned to the client.

Question 7

Complete

What is the main purpose of ARP?

a. To determine whether a received frame contains transmission errors.

b. To determine the best network path used to reach a remote destination.

c. To determine a link-layer address when the corresponding IP address is known.

d. To assign an IP address to a host joining the local network.

e. To determine an application port number used by a destination process.

Question 8

Complete

Choose an appropriate statement about ARP protocol.

- ARP request: It is normally broadcast to every device in the local network.
- ARP reply: It is normally sent directly to the host that issued the ARP Reply.

Question 9

Complete

Alice is connected to Router R1 on one LAN. Router R1 is connected to Router R2 through another network link. Bob is connected to Router R2 on a different LAN. Alice wants to send an IP packet to Bob.

Which statement about ARP across multiple networks is **NOT correct**?

a. A sender uses the MAC address of the next hop on the current link.

b. Alice must learn Bob's MAC address directly before sending the first frame.

c. Different links may use different source and destination MAC addresses.

d. Router R1 may use ARP to learn the MAC address of Router R2.

e. Router R2 may use ARP to learn Bob's MAC address on Bob's local network.

Question 10

Complete

Which statement about RARP is **NOT correct**?

a. RARP was mainly associated with legacy devices such as diskless workstations.

b. RARP can help a host obtain its IP address from a known MAC address.

c. A RARP server checks its table to find the IP address matching the MAC address.

d. RARP can provide a subnet mask, default gateway, and other network settings automatically.

e. RARP normally sends a broadcast request to locate a RARP server on the network.

Question 11

Complete

Which protocol/function mapping is **NOT correct**?

a. ARP — broadcast request on the local link

b. ARP — obtain MAC address from a known IP address

c. RARP — obtain IP address using MAC-related information

d. DHCP — obtain IP address and network configuration

e. DHCP — determine the destination MAC address of the default gateway